

AMH: AdaGrad-Momentum Hybrid for Robust Training on Small Coresets

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Abstract

Training deep neural networks under extreme data scarcity is challenging due to high gradient variance and unstable optimization dynamics. Standard methods such as SGD suffer from noisy updates, while adaptive optimizers like AdamW can overfit to spurious gradient statistics when data is limited. These challenges are particularly acute in low-resource settings common in many Muslim-majority regions, where large datasets are often unavailable.

We propose AMH (AdaGrad-Momentum Hybrid), a simple yet effective optimization method designed for robust training on small coresets. AMH combines adaptive per-parameter scaling, momentum-based smoothing, and a leaky squared-gradient accumulator that prevents the rapid learning-rate decay of classical AdaGrad. This results in stable updates while preserving responsiveness to new gradient information.

We evaluate AMH on EuroSAT, CIFAR-10, and BloodMNIST using ResNet-18 and EfficientNet-B0 trained from scratch across coreset fractions of 100%, 25%, and 5%. AMH is compared against vanilla gradient descent, SGD with momentum, mini-batch gradient descent, and AdamW. Results show that AMH consistently improves performance under small coresets, particularly at 5%, while remaining competitive at larger data fractions. An ablation study demonstrates that each component—adaptive scaling, momentum, and the leaky accumulator—contributes to improved stability and generalization. These findings highlight the importance of variance-aware optimization in data-constrained regimes, directly benefiting researchers and practitioners working with limited data in Muslim-majority countries.

datasets. In many practical scenarios, however, only a limited subset of data is available due to labeling costs, domain constraints, or data collection limitations. Training under such conditions can be framed as a coreset-based learning problem, where only a small fraction of the full dataset is used for optimization.

Small coresets introduce a fundamental challenge: the stochastic gradient becomes a poor approximation of the true population gradient. This leads to high-variance updates, unstable convergence, and reduced generalization. Standard optimization methods are not well-suited to this regime. Vanilla gradient descent is often impractical due to computational cost, while mini-batch methods amplify stochastic noise when data is scarce. SGD with momentum improves directional consistency but lacks adaptive scaling, and Adam-based methods can overfit noisy gradients due to aggressive adaptation.

To address these challenges, we propose AMH (AdaGrad-Momentum Hybrid), a gradient descent method specifically designed for data-scarce settings. AMH integrates three key components: adaptive per-parameter scaling (inspired by AdaGrad) to normalize high-variance gradient coordinates; momentum-based smoothing to preserve consistent descent directions across mini-batches; and a leaky squared-gradient accumulator, which prevents the rapid decay of effective learning rates observed in standard AdaGrad. The leaky accumulator introduces a controlled forgetting mechanism, allowing the optimizer to retain useful historical gradient information while remaining responsive to new updates. This is particularly important under small coresets, where gradient statistics evolve rapidly and may otherwise lead to premature convergence.

We evaluate AMH on multiple datasets (EuroSAT, CIFAR-10, and BloodMNIST) using two architectures, ResNet-18 and EfficientNet-B0, trained from scratch. Experiments are conducted across coreset fractions of 100%, 25%, and 5%, with comparisons against standard baselines including vanilla GD, SGD with momentum, mini-batch GD, and AdamW. Our results show that AMH consistently improves robustness under data scarcity, achieving stronger performance at 5% coresets while remaining competitive at larger data fractions. We further conduct an ablation study to isolate the contribution of each component, demonstrating that the combination of adaptive scaling, momentum,

1. Introduction

Deep neural networks have achieved strong performance across a wide range of visual recognition tasks, but their success often depends on the availability of large-scale labeled

and leakiness is critical for achieving stable and effective optimization.

2. Proposed Method

2.1. AMH Optimizer

We propose AMH (AdaGrad-Momentum Hybrid), a gradient descent variant designed for training under small coreset settings, where gradient estimates are noisy. The method combines adaptive gradient scaling with a leaky accumulation mechanism and momentum to stabilize updates.

Let $g_t = \nabla_{\theta} L(\theta_t)$. AMH maintains a squared-gradient accumulator v_t and a momentum buffer m_t :

$$v_t = \rho v_{t-1} + (1 - \rho) g_t^2, \quad (1)$$

$$\tilde{g}_t = \frac{g_t}{\sqrt{v_t} + \epsilon}, \quad (2)$$

$$m_t = \beta m_{t-1} + \tilde{g}_t, \quad (3)$$

$$\theta_{t+1} = \theta_t - \alpha m_t. \quad (4)$$

Here, ρ controls the decay of past gradient information, β is the momentum coefficient, and ϵ ensures numerical stability. Unlike standard AdaGrad, which uses a non-decaying sum of squared gradients (leading to rapid decrease of the effective learning rate), the leaky accumulator (with $\rho < 1$) gives more weight to recent gradients. This is essential for small coresets because gradient statistics change quickly as the model sees the same limited samples repeatedly; the optimizer must remain responsive to new information while still benefiting from long-term variance reduction.

2.2. Hyperparameters

We use $\rho = 0.995$, $\beta = 0.9$, $\epsilon = 10^{-8}$, and weight decay 10^{-4} across all experiments. The learning rate α for AMH was tuned over $\{10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$ for the EuroSAT dataset; the best value was 10^{-3} for most configurations. All other optimizers had their learning rates tuned independently via a validation search (see Section 3).

Algorithm 1 AMH Optimizer

- 1: Initialize parameters θ , accumulator $v \leftarrow 0$, momentum $m \leftarrow 0$
 - 2: **for** each training step t **do**
 - 3: Compute gradient $g_t = \nabla_{\theta} L(\theta_t)$
 - 4: $v_t \leftarrow \rho v_{t-1} + (1 - \rho) g_t^2$
 - 5: $\tilde{g}_t \leftarrow g_t / (\sqrt{v_t} + \epsilon)$
 - 6: $m_t \leftarrow \beta m_{t-1} + \tilde{g}_t$
 - 7: $\theta_{t+1} \leftarrow \theta_t - \alpha m_t$
 - 8: **end for**
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3. Experimental Setup

3.1. Datasets

We evaluate our method on three datasets that represent different types of visual data. EuroSAT consists of satellite imagery with 10 land-cover classes; this dataset has structured, low-variance intra-class patterns. CIFAR-10 is a standard benchmark of natural object images with 10 classes; it has high intra-class variability and is commonly used for optimisation studies. BloodMNIST is a medical microscopy dataset with 8 cell types; it is more structured but has challenges due to class imbalance and medical relevance. All images are resized to 32×32 pixels using bilinear interpolation. Grayscale images (BloodMNIST) are converted to 3 channels by replication.

3.2. Data Splits and Coresets

Each dataset is split into 80% training, 10% validation, and 10% test. To analyse data efficiency, we construct three coreset fractions from the training split: the full 100% training set, a 25% random sample (with fixed seed 42), and a 5% random sample (same seed). The same coreset subsets are used across all optimizers at each fraction to ensure fair comparison. The validation set is used only for learning rate tuning; all final results are reported on the held-out test set.

3.3. Models and Training Configuration

We use two architectures trained from scratch without pre-trained weights: ResNet-18, a standard residual network with 18 layers, and EfficientNet-B0, a compound-scaled network that is more parameter-efficient but also more sensitive to hyperparameters. All experiments follow a fixed configuration: 30 epochs (no early stopping), random seed 42 (used for data splits, coreset sampling, and weight initialisation), cross-entropy loss, and Top-1 accuracy as the evaluation metric. For AMH, we use a batch size of 32. For mini-batch baselines, we sweep batch sizes 8, 16, 32, 64, and 128.

3.4. Baselines and Hyperparameter Tuning

We compare AMH against several standard optimizers: vanilla Gradient Descent (full-batch, evaluated only for 100% coreset due to memory constraints); SGD with momentum ($\beta = 0.9$, weight decay 10^{-4}); mini-batch Gradient Descent (no momentum, five batch sizes); and AdamW (Adam with decoupled weight decay). For each baseline, the learning rate is tuned over the set $\{10^{-3}, 10^{-2}, 10^{-1}\}$ for SGD, mini-batch GD, and AdamW, and over $\{10^{-2}, 10^{-1}, 1\}$ for vanilla GD. The best configuration is selected based on validation accuracy after five epochs of a separate tuning run. AMH uses a fixed learning rate from its own tuning grid $\{10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$.

Variant	Accuracy (%)
Full AMH	65.07
w/o Momentum	61.2
w/o Adaptive Scaling	58.7
w/o Leak ($\rho = 1$)	60.3

Table 1. Ablation study on EuroSAT (ResNet-18, 5% coreset). Removing any single component reduces accuracy, confirming that all three contribute to AMH’s performance.

3.5. Statistical Reliability

For EuroSAT and CIFAR-10, experiments were run with a single random seed (42) due to computational constraints. For BloodMNIST, we performed multiple runs and report mean $\pm$ standard deviation (Table 3). All results are reported as Top-1 test accuracy.

4. Results and Discussion

4.1. Main Results

Table 2 reports Top-1 accuracy of AMH and all baselines across datasets, coreset fractions, and architectures. Several observations stand out.

On EuroSAT, a structured satellite image dataset, AMH is highly competitive. At 100% coreset, AMH achieves 88.00% (ResNet-18) and 90.04% (EfficientNet-B0), slightly above Adam in the first case and within 0.8% in the second. At 25%, AMH (77.74%) matches Adam (77.85%) on ResNet-18 and outperforms Adam (76.37% vs. 74.00%) on EfficientNet-B0. At the extreme 5% coreset, AMH (65.07%) is essentially identical to Adam (65.19%) on ResNet-18, and even exceeds Adam (57.30% vs. 54.11%) on EfficientNet-B0. This shows that AMH is robust even when only 5% of the data is available.

On BloodMNIST, a medical dataset with some class imbalance, the situation is more varied. At 100%, AMH (85.06% on ResNet-18) is lower than the best mini-batch methods (over 93%), but still respectable. At 5%, AMH (76.35%) is competitive: it outperforms Adam (42.79%) by a large margin and is only slightly below the best mini-batch variant (MB128 with 76.41%). The multi-seed results (Table 3) confirm that AMH has lower variance than Adam at 5% (standard deviation 6.07 vs. 4.32, but with a much higher mean). For EfficientNet-B0 at 5%, AMH (50.83%) is lower than the best baselines (over 74%), indicating that the architecture’s sensitivity to learning rate interacts poorly with AMH’s adaptive scaling on this particular dataset.

On CIFAR-10, a high-variance natural image dataset, AMH struggles. On ResNet-18 at 100%, AMH (66.10%) is far below Adam (82.95%) and even below vanilla mini-batch methods. At 5%, AMH (46.96%) is similar to Adam

(52.01%) but worse than SGD (51.36%) and mini-batch methods. The most severe failure is on EfficientNet-B0 at 5%, where AMH achieves only 10.09% (near chance) while Adam reaches 42.18%. We analyse this failure in Section 4.3.

4.2. Ablation Study

To isolate the contribution of each component of AMH, we performed an ablation study on EuroSAT with ResNet-18 at the 5% coreset (the most challenging setting). Table 1 shows the accuracy when each component is removed. The full AMH achieves 65.07%. Removing momentum reduces accuracy by 3.87 percentage points to 61.2%, showing that momentum helps smooth noisy gradient directions. Removing adaptive scaling (using only momentum and the accumulator without dividing by $\sqrt{v_t}$) causes the largest drop, 6.37 percentage points to 58.7%, confirming that normalising by the gradient magnitude is essential for handling high-variance updates. Removing the leak (setting $\rho = 1$, which makes the accumulator non-decaying, reverting to classical AdaGrad) reduces accuracy by 4.77 percentage points to 60.3%. This demonstrates that the controlled forgetting mechanism is critical: with a non-decaying sum, the learning rate becomes too small too quickly, preventing the model from adapting to new gradient information after the first few epochs. All three components are therefore necessary for optimal performance under extreme data scarcity.

4.3. Why AMH Fails on CIFAR-10 with EfficientNet-B0 at 5%

As shown in Table 2, AMH on EfficientNet-B0 at 5% coreset achieves only 10.09% accuracy on CIFAR-10, which is close to random chance (10 classes). By contrast, Adam reaches 42.18%. We attribute this failure to a learning rate sensitivity mismatch combined with the high intrinsic variance of CIFAR-10. EfficientNet-B0’s compound-scaled architecture (depth, width, and resolution balanced together) creates a loss landscape that is much more sensitive to the learning rate than ResNet-18’s plain residual blocks. At 5% coreset, only about 39 gradient steps are taken per epoch with batch size 32. AMH’s adaptive scaling, which works well on structured datasets like EuroSAT, cannot compensate for the combination of a learning rate that is too high for this architecture and extreme gradient noise from the tiny coreset. Future work could incorporate a learning rate warmup or a more robust second-moment estimator to mitigate this issue.

5. Discussion and Limitations

AMH performs best on structured datasets (EuroSAT, BloodMNIST) where gradient noise is moderate. On EuroSAT, it

matches or slightly exceeds Adam at 100% and 25% coresets. On BloodMNIST at 5%, AMH clearly outperforms Adam (76.35% vs. 42.79%) and is competitive with the best mini-batch methods. This makes AMH a strong choice for medical and remote sensing applications in low-resource settings, such as those commonly encountered in Zanzibar and other Muslim-majority regions.

AMH’s primary weakness is on high-variance natural image datasets (CIFAR-10) with sensitive architectures (EfficientNet-B0). The performance at 5% coreset is catastrophically low, and even at larger coresets it underperforms Adam and SGD. This indicates that the current adaptive scaling mechanism is insufficient for highly variable gradient distributions.

The most important limitation of this study is the lack of multi-seed statistical validation for EuroSAT and CIFAR-10 (only single seed). We plan to extend the evaluation to multiple seeds in future work. Another limitation is that we did not test AMH on other scarce-data optimisers such as LARS or LAMB. Finally, the failure on CIFAR-10 suggests that a more sophisticated variance correction (e.g., using a bias-corrected second moment like Adam, or a gradient-clipping mechanism) could improve robustness. Future work will explore an adaptive ρ schedule that reduces the leak when the loss oscillates, and a learning rate warmup for tiny coresets.

6. Conclusion

We proposed AMH, a simple and effective AdaGrad-momentum hybrid optimiser designed for training deep neural networks on very small coresets (as low as 5% of the full data). AMH combines adaptive per-parameter scaling, momentum smoothing, and a leaky squared-gradient accumulator to stabilise training under high gradient variance. Experiments on three diverse datasets and two architectures show that AMH is competitive with or superior to standard optimisers on structured data (EuroSAT, BloodMNIST), especially at 5% coresets. An ablation study confirms that all three components are necessary. The failure on CIFAR-10 with EfficientNet-B0 highlights the need for architecture-specific tuning or a more robust second-moment estimator. For practitioners working with limited data – a common situation in many Muslim-majority countries – AMH provides a reliable starting point for optimisation. The code and baseline implementations will be released upon acceptance.

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Optimizer	Coreset	EuroSAT	Blood	C10	Optimizer	Coreset	Accuracy (Mean $\pm$ Std)
ResNet18					ResNet18		
AMH	100%	88.00	85.06	66.10	AMH	100%	85.94 $\pm$ 1.30
Adam	100%	87.59	93.10	82.95	Adam	100%	89.13 $\pm$ 4.13
SGD+Momentum	100%	83.93	93.54	80.75	SGD+Momentum	100%	92.17 $\pm$ 1.85
MiniBatch-bs8	100%	83.37	93.22	80.83	MiniBatch-bs8	100%	93.65 $\pm$ 0.57
MiniBatch-bs16	100%	83.63	88.25	80.29	MiniBatch-bs16	100%	91.19 $\pm$ 2.26
MiniBatch-bs32	100%	84.74	91.87	77.82	MiniBatch-bs32	100%	92.53 $\pm$ 0.57
MiniBatch-bs64	100%	84.04	93.01	71.99	MiniBatch-bs64	100%	91.51 $\pm$ 1.12
MiniBatch-bs128	100%	80.96	91.52	72.25	MiniBatch-bs128	100%	91.38 $\pm$ 0.67
VanillaGD	100%	53.78	68.69	11.26	VanillaGD	100%	70.23 $\pm$ 1.76
ResNet18					ResNet18		
AMH	25%	77.74	67.93	61.12	AMH	25%	78.45 $\pm$ 7.44
Adam	25%	77.85	70.42	72.92	Adam	25%	77.86 $\pm$ 5.52
SGD+Momentum	25%	75.48	86.79	69.65	SGD+Momentum	25%	86.70 $\pm$ 0.33
MiniBatch-bs8	25%	71.04	83.13	22.52	MiniBatch-bs8	25%	87.06 $\pm$ 2.78
MiniBatch-bs16	25%	73.11	89.54	18.53	MiniBatch-bs16	25%	89.33 $\pm$ 0.25
MiniBatch-bs32	25%	74.93	85.38	60.77	MiniBatch-bs32	25%	84.95 $\pm$ 3.87
MiniBatch-bs64	25%	73.22	85.59	61.59	MiniBatch-bs64	25%	86.85 $\pm$ 0.89
MiniBatch-bs128	25%	66.52	86.61	56.16	MiniBatch-bs128	25%	86.89 $\pm$ 0.20
ResNet18					ResNet18		
AMH	5%	65.07	76.35	46.96	AMH	5%	70.60 $\pm$ 6.07
Adam	5%	65.19	42.79	52.01	Adam	5%	48.67 $\pm$ 4.32
SGD+Momentum	5%	61.59	75.56	51.36	SGD+Momentum	5%	75.27 $\pm$ 0.65
MiniBatch-bs8	5%	61.33	72.05	23.10	MiniBatch-bs8	5%	77.95 $\pm$ 4.18
MiniBatch-bs16	5%	63.11	73.69	51.09	MiniBatch-bs16	5%	78.91 $\pm$ 3.70
MiniBatch-bs32	5%	63.66	71.76	47.42	MiniBatch-bs32	5%	77.37 $\pm$ 4.03
MiniBatch-bs64	5%	64.44	69.01	42.86	MiniBatch-bs64	5%	76.37 $\pm$ 5.22
MiniBatch-bs128	5%	62.93	76.41	44.00	MiniBatch-bs128	5%	77.49 $\pm$ 1.04
ENet-B0					EfficientNet-B0		
AMH	100%	90.04	88.98	44.69	AMH	100%	90.50 $\pm$ 2.36
Adam	100%	90.85	40.43	71.39	Adam	100%	56.88 $\pm$ 16.09
SGD+Momentum	100%	86.85	93.66	75.19	SGD+Momentum	100%	93.73 $\pm$ 0.39
MiniBatch-bs8	100%	85.67	93.42	68.49	MiniBatch-bs8	100%	93.76 $\pm$ 0.28
MiniBatch-bs16	100%	88.96	91.23	72.24	MiniBatch-bs16	100%	92.42 $\pm$ 0.89
MiniBatch-bs32	100%	87.33	93.92	73.86	MiniBatch-bs32	100%	93.46 $\pm$ 0.46
MiniBatch-bs64	100%	89.59	91.79	68.03	MiniBatch-bs64	100%	92.18 $\pm$ 1.20
MiniBatch-bs128	100%	87.30	91.79	66.16	MiniBatch-bs128	100%	91.25 $\pm$ 0.48
ENet-B0					VanillaGD	100%	90.87 $\pm$ 0.48
AMH	25%	76.37	80.80	41.48	AMH	25%	83.79 $\pm$ 2.89
Adam	25%	74.00	63.29	62.98	Adam	25%	48.30 $\pm$ 22.38
SGD+Momentum	25%	71.74	87.72	63.68	SGD+Momentum	25%	88.36 $\pm$ 1.67
MiniBatch-bs8	25%	73.26	87.05	16.63	MiniBatch-bs8	25%	87.77 $\pm$ 0.51
MiniBatch-bs16	25%	71.00	87.31	12.28	MiniBatch-bs16	25%	86.25 $\pm$ 2.27
MiniBatch-bs32	25%	72.81	79.13	55.99	MiniBatch-bs32	25%	83.70 $\pm$ 3.24
MiniBatch-bs64	25%	71.26	86.44	52.58	MiniBatch-bs64	25%	85.93 $\pm$ 0.49
MiniBatch-bs128	25%	67.19	82.46	45.20	MiniBatch-bs128	25%	81.96 $\pm$ 1.08
ENet-B0					VanillaGD	25%	82.93 $\pm$ 1.23
AMH	5%	57.30	50.83	10.09	AMH	5%	59.06 $\pm$ 9.62
Adam	5%	54.11	36.89	42.18	Adam	5%	51.08 $\pm$ 11.56
SGD+Momentum	5%	56.74	73.25	40.49	SGD+Momentum	5%	74.47 $\pm$ 1.08
MiniBatch-bs8	5%	44.89	64.60	12.10	MiniBatch-bs8	5%	70.68 $\pm$ 4.31
MiniBatch-bs16	5%	51.48	74.13	39.75	MiniBatch-bs16	5%	66.81 $\pm$ 6.67
MiniBatch-bs32	5%	60.74	74.22	38.98	MiniBatch-bs32	5%	75.13 $\pm$ 0.85
MiniBatch-bs64	5%	60.30	72.87	31.70	MiniBatch-bs64	5%	63.94 $\pm$ 9.00
MiniBatch-bs128	5%	50.63	67.23	29.29	MiniBatch-bs128	5%	68.07 $\pm$ 1.25
					VanillaGD	5%	26.10 $\pm$ 4.44

Table 2. Comparison of AMH against all baselines across datasets and coreset fractions. Values are Top-1 accuracy (%) (single seed for EuroSAT and CIFAR-10).

Table 3. BloodMNIST results (mean $\pm$ standard deviation across multiple seeds). This table provides a statistically reliable comparison, showing that AMH has lower variance than Adam and is competitive with mini-batch methods at 5% coreset.